



HOW TO USE THIS SET

20 questions, 5 options each (A–E). Answer on paper or in your head, then check every answer against the key on the final page.

4 Easy · 10 Medium · 6 Hard · suggested time ≈ 25 minutes.

Q Questions

1. What do the epithelial cells of the large intestine secrete, aside from inorganic substances?

EASY

- A. A full complement of digestive enzymes
- B. Mucus only
- C. Bile acids
- D. Pepsinogen
- E. Intrinsic factor

2. Which colonic movement drives contents into the rectum 3–4 times per day, typically triggered by the gastrocolic reflex?

EASY

- A. Haustral churning
- B. Segmentation
- C. Mass peristalsis
- D. Standard peristalsis only
- E. Retroperistalsis

3. Which portion of the colon is primarily responsible for water and electrolyte absorption, functioning as the "absorbing colon"?

EASY

- A. Distal colon
- B. Rectum
- C. Proximal colon
- D. Anal canal
- E. Sigmoid colon exclusively

4. What gives feces their characteristic brown color?

EASY

- A. Unabsorbed bile salts
- B. Stercobilin, from bacterial breakdown of bilirubin
- C. Excess mucus
- D. Sloughed epithelial cells
- E. Dietary fiber alone

5. A patient reports passing ropy mucus every 30 minutes with very little fecal material. Which physiological state best explains this?

MEDIUM

- A. Failure of the ileocecal sphincter
- B. Extreme parasympathetic stimulation of the colon, which increases mucus secretion and peristaltic motility
- C. Complete loss of haustral churning
- D. Isolated bacterial overgrowth in the cecum
- E. Loss of mass peristalsis entirely

6. A patient with obstructive jaundice (bile duct obstruction) develops pale, clay-colored stools. Which mechanism best explains this finding?

MEDIUM

- A. Reduced bile salt delivery impairing fat digestion only, with no color change expected
- B. Reduced bilirubin delivery to the colon, reducing bacterial conversion to stercobilin, the pigment responsible for normal brown stool color
- C. Increased mucus secretion diluting stool color
- D. Excess vitamin K absorption
- E. Increased haustral churning bleaching the stool

7. A patient undergoes a procedure that eliminates the gastrocolic reflex pathway. Which colonic motor pattern would be most directly and specifically reduced?

MEDIUM

- A. Haustral churning
- B. Segmentation in the small intestine
- C. Mass peristalsis
- D. Ileocecal sphincter tone at rest
- E. Standard slow peristalsis

8. A patient with chronic constipation is found to have particularly sluggish activity in the distal colon and rectum, with normal proximal colonic function. Which functional description best matches the affected region?

MEDIUM

- A. The "absorbing colon," responsible for most water/electrolyte uptake
- B. The "storage colon," which holds feces until defecation
- C. The region responsible for bacterial vitamin K synthesis
- D. The region where most carbohydrate fermentation occurs
- E. The region responsible for ileocecal sphincter regulation

9. A patient on long-term broad-spectrum antibiotics that eliminate colonic bacterial flora develops unexpected bruising and abnormal coagulation studies. Which colonic bacterial function loss most directly explains this?

MEDIUM

- A. Loss of bacterial mucin production
- B. Loss of bacterial production of vitamin K, which contributes to blood clotting
- C. Loss of bacterial water absorption
- D. Loss of bacterial bicarbonate secretion
- E. Loss of bacterial fermentation of carbohydrates only

10. A patient consumes a large amount of a poorly digestible carbohydrate (e.g., certain fibers) and develops excessive flatus. Which colonic process directly explains this symptom?

MEDIUM

- A. Bacterial fermentation of remaining carbohydrate, producing H₂, CO₂, and methane gases
- B. Increased mass peristalsis compressing luminal air
- C. Excess bile salt reabsorption
- D. Increased mucus secretion trapping air
- E. Excess water absorption concentrating gas

11. A patient with ileocecal valve incompetence (the valve fails to close properly) after surgery is most likely to show which downstream effect?

MEDIUM

- A. Excessive retention of chyme in the ileum
- B. Reflux of colonic bacterial content back into the relatively sterile ileum, risking bacterial overgrowth
- C. Increased vitamin B₁₂ absorption
- D. Improved water absorption in the proximal colon
- E. No physiological consequence, since the valve has no bacteriological function

12. A patient's stool analysis shows sloughed epithelial cells, bacteria, unabsorbed digested material, and inorganic salts, consistent with normal feces formation. Roughly what fraction of the 0.5-1.0 L of water entering the colon daily remains unabsorbed in normal physiology?

MEDIUM

- A. ~50%
- B. ~10-20%
- C. ~100-200 mL (about 10-20% of the total, i.e. a small remainder)
- D. Essentially all of it
- E. None — the colon absorbs 100% of water entering it

13. A patient develops a colonic segment where indole and skatole (bacterial protein breakdown products) fail to be absorbed and detoxified, instead accumulating and being excreted almost entirely in feces rather than partly in urine. What does this reveal about the normal handling of these compounds?

MEDIUM

- A. Indole and skatole are normally partly absorbed, detoxified by the liver, and excreted in urine, with the rest eliminated in feces (contributing to odor) — so this patient's pattern reflects a deviation from the normal split
- B. Indole and skatole are always excreted entirely in feces under normal circumstances, so this is not a deviation at all
- C. These compounds are normally produced only by the small intestine, not the colon
- D. These compounds have no relationship to bacterial protein metabolism
- E. These compounds are entirely reabsorbed and never appear in feces normally

14. A patient with excessive parasympathetic tone due to a pelvic nerve irritation shows both increased mucus secretion and increased peristaltic motility of the colon. What does this pairing best illustrate about pelvic parasympathetic innervation of the colon?

MEDIUM

- A. Parasympathetic stimulation affects only secretion, not motility
- B. Parasympathetic stimulation affects only motility, not secretion
- C. Pelvic (parasympathetic) nerve stimulation increases both mucus secretion and peristaltic motility together, reflecting a combined secretomotor effect
- D. Parasympathetic stimulation has no effect on either process
- E. This pairing indicates sympathetic, not parasympathetic, involvement

15. A patient has a colonic resection removing the proximal (ascending and transverse) colon entirely, with the distal colon and rectum preserved and re-anastomosed to the terminal ileum. Predict the most direct consequence for water and electrolyte handling.

HARD

- A. No consequence, since the distal colon fully compensates for all absorptive functions
- B. Substantially reduced water/electrolyte absorptive capacity, since the proximal colon (the "absorbing colon") has been removed, likely producing looser, higher-volume stool even though the distal "storage colon" remains
- C. Improved water absorption, since the distal colon absorbs more efficiently than the proximal colon
- D. Complete cessation of defecation, since the proximal colon is required to trigger the defecation reflex
- E. No change in stool consistency, since fecal water content is determined entirely in the rectum

16. A researcher compares two patients: one with loss of haustral churning but preserved mass peristalsis, and another with the reverse (preserved haustral churning but loss of mass peristalsis via gastrocolic reflex disruption). Which prediction about their bowel patterns is most defensible?

HARD

- A. Both patients would have identical bowel patterns, since the two movements are functionally redundant
- B. The first patient would have impaired local mixing/absorption optimization but might still have periodic bulk transit via mass peristalsis; the second patient would have preserved local mixing but impaired triggered bulk transit into the rectum, more likely presenting as timing-related difficulty with defecation
- C. Loss of either movement alone would cause complete colonic obstruction
- D. Mass peristalsis is unrelated to any reflex and would be unaffected by gastrocolic reflex disruption
- E. Haustral churning is solely responsible for moving contents into the rectum, making the second patient's pattern impossible

17. A patient with a colostomy (external opening from the ascending colon, bypassing the rest of the colon) has consistently liquid, high-volume ostomy output despite normal small intestinal function. Which combination of missing colonic functions best explains this, considering the segments now bypassed?

HARD

- A. Loss of gastric acid secretion, since the colon has no bearing on stool consistency
- B. Loss of the transverse, descending, and sigmoid colon's water/electrolyte absorption and storage function, since the ascending colon alone (or its most proximal portion) has not yet completed the bulk of colonic water reabsorption and has no distal storage segment to further concentrate and hold the stool
- C. Loss of bacterial vitamin K synthesis exclusively, since this alone controls stool consistency
- D. Loss of gastric emptying regulation, unrelated to the colon at all
- E. Improved absorption due to a shorter transit path

18. A patient with chronic use of a stimulant laxative that increases colonic motility develops electrolyte disturbances, including hypokalemia. Considering normal colonic ion handling, which mechanism most directly explains this?

HARD

- A. Excessive haustral churning destroys potassium directly
- B. Accelerated transit reduces contact time for normal colonic absorption, while ongoing potassium loss (normally absorbed but also secreted/lost in stool) is amplified by the higher stool volume and faster flow, net increasing fecal potassium loss
- C. Laxatives block bacterial vitamin K synthesis, which directly causes hypokalemia
- D. Colonic motility has no relationship to electrolyte handling
- E. This is caused entirely by the small intestine, not the colon

19. A physiologist notes that under normal conditions, the large intestine performs almost no enzymatic digestion, yet substantial "digestion" of remaining nutrients still occurs there via bacteria. A patient on antibiotics that eliminate colonic flora would be expected to show which change in stool composition?

HARD

- A. No change at all, since the colon's minimal digestive role means bacterial elimination is inconsequential
- B. Increased undigested/unfermented carbohydrate and protein breakdown products in stool, since bacterial fermentation (the colon's primary "digestive" process) is lost, alongside reduced production of bacterial-dependent vitamins and pigments like stercobilin
- C. Complete cessation of stool formation entirely
- D. Markedly increased stercobilin production
- E. No effect on flatus production, since gas is generated by host enzymes, not bacteria

20. A patient has an unusual congenital anomaly in which the crypts of Lieberkühn in the colon are absent, but all other colonic structures (musculature, ileocecal valve, rectum) are normal. Predict the earliest and most direct physiological consequence.

HARD

- A. Complete loss of water absorption, since crypts are the primary water-absorbing structures
- B. Loss of colonic mucus secretion, since crypts of Lieberkühn are the source of colonic mucus, likely producing a dry, poorly lubricated stool prone to mucosal injury and slowed transit
- C. No consequence, since crypts of Lieberkühn play no functional role in the colon
- D. Loss of mass peristalsis specifically, since it depends on crypt secretion
- E. Complete loss of bacterial fermentation capacity

✓ Answer key

Q	Answer	Difficulty	Why
1	B	EASY	The large intestine secretes mucus (and inorganic substances) only — it has almost no digestive enzymes.
2	C	EASY	Mass peristalsis, triggered by the gastrocolic reflex, drives colonic contents into the rectum 3–4 times daily.
3	C	EASY	The proximal colon is the "absorbing colon"; the distal colon mainly stores feces.
4	B	EASY	Bacterial decomposition of bilirubin produces stercobilin, giving feces their brown color.
5	B	MEDIUM	Extreme parasympathetic (pelvic nerve) stimulation increases both mucus secretion and motility, producing frequent ropy mucus discharge with little fecal content.
6	B	MEDIUM	Biliary obstruction reduces bilirubin delivery to the colon; less bacterial conversion to stercobilin results in pale, clay-colored stools.

7	C	MEDIUM	Mass peristalsis is specifically triggered by the gastrocolic reflex (food entering the stomach); its pathway loss would most directly reduce this movement.
8	B	MEDIUM	The distal colon functions as the "storage colon," holding feces until defecation; dysfunction here is consistent with distal-predominant constipation.
9	B	MEDIUM	Colonic bacteria produce vitamin K (along with B vitamins); their elimination by antibiotics can contribute to vitamin K deficiency and coagulopathy.
10	A	MEDIUM	Colonic bacteria ferment unabsorbed carbohydrate into gases (H ₂ , CO ₂ , methane), producing flatus, which becomes excessive with a large fermentable carbohydrate load.
11	B	MEDIUM	The ileocecal valve normally limits reflux of colonic (bacteria-rich) contents into the ileum; incompetence allows bacterial reflux and risks small intestinal bacterial overgrowth.
12	C	MEDIUM	Of the 0.5–1.0 L of water entering the colon daily, only about 100–200 mL is not absorbed (the rest returns to the body via osmosis).
13	A	MEDIUM	Normally, bacteria convert protein breakdown products into indole/skatole; some is eliminated in feces (contributing to odor) while the rest is absorbed, detoxified by the liver, and excreted in urine — the described patient shows a deviation from this normal dual-route handling.
14	C	MEDIUM	Pelvic nerve (parasympathetic) stimulation increases both mucus secretion and peristaltic motility of the colon simultaneously, a combined secretomotor effect.
15	B	HARD	Because the proximal colon does most of the water/electrolyte absorption ("absorbing colon"), its removal — even with the distal "storage colon" preserved — would be expected to reduce absorptive capacity and produce looser stool, since storage function alone cannot substitute for absorption.
16	B	HARD	Haustral churning performs continuous local mixing/absorption optimization, while mass peristalsis provides bulk, meal-triggered transit toward the rectum; losing one selectively would produce distinguishable functional patterns rather than identical or catastrophic outcomes.
17	B	HARD	An ascending colostomy bypasses the transverse, descending, and sigmoid colon and rectum — segments that complete water/electrolyte absorption and provide storage — so effluent remains liquid and high-volume compared to output from a more distal colostomy or normal defecation.
18	B	HARD	Faster colonic transit shortens the time available for normal water/electrolyte reabsorption; combined with higher stool water/volume, this increases net fecal loss of potassium and other electrolytes, explaining laxative-associated hypokalemia.
19	B	HARD	Since colonic "digestion" is bacterial rather than enzymatic, eliminating flora removes fermentation of residual carbohydrate/protein and bacterial synthesis of vitamins and stercobilin, changing stool composition, odor, color, and flatus production.
20	B	HARD	The crypts of Lieberkühn in the colon are the source of mucus secretion (the colon's main secretory product); their absence would primarily impair lubrication/protection of the mucosa rather than water absorption (a surface epithelial function) or motility.